

● MEDIUM

● ALGEBRA

● ANSWER KEY

Algebra

17

10 answers · Rationale-rich · Teach from doc

Q1**B**

B. $fx = -2x + 7$

Choice B is correct. The graph of a line in the xy -plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the y -intercept. The slope of a line that passes through the points x_1, y_1 and x_2, y_2 can be calculated using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $-1, -2$ and $0, -4$. Substituting $-1, -2$ and $0, -4$ for x_1, y_1 and x_2, y_2 , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{-4 - (-2)}{0 - (-1)}$, which is equivalent to $m = \frac{-2}{1}$, or $m = -2$. Since the line shown passes through the point $0, -4$, it follows that $b = -4$. Substituting -2 for m and -4 for b in the equation $y = mx + b$ yields $y = -2x - 4$. It's given that the graph shown is the graph of $y = fx - 11$. Substituting $-2x - 4$ for y in the equation $y = fx - 11$ yields $-2x - 4 = fx - 11$. Adding 11 to both sides of this equation yields $-2x + 7 = fx$. Therefore, the equation $fx = -2x + 7$ defines the linear function f .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**C**

C. $-2, -5$

Choice C is correct. Adding $9x$ to both sides of the first equation in the given system yields $y = 9x + 13$. Substituting the expression $9x + 13$ for y in the second equation in the given system yields $5x = 29x + 13$. Distributing the 2 on the right-hand side of this equation yields $5x = 18x + 26$. Subtracting $18x$ from both sides of this equation yields $-13x = 26$. Dividing both sides of this equation by -13 yields $x = -2$. Substituting -2 for x in the equation $y = 9x + 13$ yields $y = 9(-2) + 13$, or $y = -5$. Therefore, the solution x, y to the given system of equations is $-2, -5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the solution y, x , not x, y , to the given system of equations.

Q3**D**

D. $y = -\frac{x}{3} + 13$

Choice D is correct. The equation that defines line t in the xy -plane can be written in slope-intercept form $y = mx + b$, where m is the slope of line t and b is its y -intercept. It's given that line t has a slope of $-\frac{1}{3}$. Therefore, $m = -\frac{1}{3}$. Substituting $-\frac{1}{3}$ for m in the equation $y = mx + b$ yields $y = -\frac{1}{3}x + b$, or $y = -\frac{x}{3} + b$. It's also given that line t passes through the point $9, 10$. Substituting 9 for x and 10 for y in the equation $y = -\frac{x}{3} + b$ yields $10 = -\frac{9}{3} + b$, or $10 = -3 + b$. Adding 3 to both sides of this equation yields $13 = b$. Substituting 13 for b in the equation $y = -\frac{x}{3} + b$ yields $y = -\frac{x}{3} + 13$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This equation defines a line that has a slope of 9 , not $-\frac{1}{3}$, and passes through the point $0, 10$, not $9, 10$.

Choice C is incorrect. This equation defines a line that passes through the point $0, 10$, not $9, 10$.

Q4**A**

A. The electrician's hourly rate

Choice A is correct. It's given that the electrician charges a onetime fee plus an hourly rate. It's also given that the graph represents the total charge, in dollars, for x hours of work. This graph shows a linear relationship in the xy -plane. Thus, the total charge y , in dollars, for x hours of work can be represented as $y = mx + b$, where m is the slope and b is the y -intercept of the graph of the equation in the xy -plane. Since the given graph represents the total charge, in dollars, by an electrician for x hours of work, it follows that its slope is m , or the electrician's hourly rate.

Choice B is incorrect. The electrician's onetime fee is represented by the y -coordinate of the y -intercept, not the slope, of the graph.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q5**25**

The correct answer is 25. Subtracting the second equation from the first equation in the given system of equations yields $2a - 2a + 8b - 4b = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by 4 yields $b = 25$.

Q6**A****A.** 5

Choice A is correct. It's given that a batch of banana milkshakes consists of 4 cups of ice cream and 2 bananas and has 1,114 mg of calcium. It's also given that there is 276 mg of calcium in 1 cup of the ice cream used to make this batch of milkshakes. It follows that the total amount of calcium in 4 cups of ice cream is $4(276)$, or 1,104 mg. Let x represent the amount of calcium, in mg, in 1 banana. It follows that the total amount of calcium in 2 bananas is $2x$ mg. Since the batch of banana milkshakes has 1,114 mg of calcium, the equation $1,104 + 2x = 1,114$ represents this situation. Subtracting 1,104 from both sides of this equation yields $2x = 10$. Dividing both sides of this equation by 2 yields $x = 5$. Therefore, the amount of calcium in 1 banana is 5 mg.

Choice B is incorrect. This is the amount of calcium, in mg, in 2 bananas, not in 1 banana.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the amount of calcium, in mg, in 4 cups of ice cream, not in 1 banana.

Q7**B****B.** $k = 4.2x$

Choice B is correct. The relationship between x food calories and k kilojoules can be modeled as a proportional relationship. Let (x_1, k_1) and (x_2, k_2) represent the values in the first two rows in the table: $(4.0, 16.7)$ and $(9.0, 37.7)$. The rate of change, or $\frac{(k_2 - k_1)}{(x_2 - x_1)}$, is $\frac{21}{5} = 4.2$; therefore, the equation that best represents the relationship between x and k is $k = 4.2x$.

Choice A is incorrect and may be the result of calculating the rate of change using $\frac{(x_2 - x_1)}{(k_2 - k_1)}$.

Choice C is incorrect because the number of kilojoules is greater than the number of food calories. Choice D is incorrect and may be the result of an error when setting up the equation.

Q8**D**

D. 6

Choice D is correct. Adding the corresponding sides of the two equations eliminates y and yields $x = 6$, as shown.

$$\begin{array}{r} \frac{1}{2}y = 4 \\ x - \frac{1}{2}y = 2 \\ \hline x + 0 = 6 \end{array}$$

If (x, y) is a solution to the system, then (x, y) satisfies both equations in the system and any equation derived from them. Therefore, $x = 6$.

Choices A, B, and C are incorrect and may be the result of errors when solving the system.

Q9**B**B. $\frac{4}{9}$

Choice B is correct. It's given that line s is parallel to line r in the xy -plane. This means that the slope of line r is equal to the slope of line s . Line r is defined by the equation $4x - 9y = 3$. This equation can be written in slope-intercept form $y = mx + b$, where m represents the slope of the line and b represents the y -coordinate of the y -intercept of the line. Subtracting $4x$ from both sides of the equation $4x - 9y = 3$ yields $-9y = -4x + 3$. Dividing both sides of this equation by -9 yields $y = \frac{4}{9}x - \frac{1}{3}$. Therefore, the slope of line r is $\frac{4}{9}$. Since parallel lines have equal slopes, the slope of line s is also $\frac{4}{9}$.

Choice A is incorrect. This is the reciprocal of the slope of line s , not the slope of line s .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10

C

C. 35

Choice C is correct. It's given that $fx = \frac{x+15}{5}$ and $fa = 10$, where a is a constant. Therefore, for the given function f , when $x = a$, $fx = 10$. Substituting a for x and 10 for fx in the given function f yields $10 = \frac{a+15}{5}$. Multiplying both sides of this equation by 5 yields $50 = a + 15$. Subtracting 15 from both sides of this equation yields $35 = a$. Therefore, the value of a is 35.

Choice A is incorrect. This is the value of a if $fa = 4$.

Choice B is incorrect. This is the value of a if $fa = 5$.

Choice D is incorrect. This is the value of a if $fa = 16$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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